

EC 441 Midterm Review: Practice Problems

Introduction to Computer Networking — Spring 2026

Boston University

These problems cover all material from Lectures 1–8 (network fundamentals through Ethernet/802.3). Work through each problem *before* consulting the hints.

Contents

1 Network Fundamentals & Delays (Lecture 1)

Problem 1.1 — End-to-End Delay

A file of 5 MB is transferred over a path with three links in series. Each link has:

- Capacity $R = 10$ Mb/s
- Length $d = 1500$ km
- Propagation speed $v = 2 \times 10^8$ m/s

The file is sent as a *single* packet (no fragmentation).

- Calculate the transmission delay on one link.
- Calculate the propagation delay on one link.
- What is the total end-to-end delay?
- Which delay dominates here, and why?

Hint

$d_{trans} = L/R$ where L is in bits. $d_{prop} = d/v$. With store-and-forward over N links: total $= N \cdot d_{trans} + N \cdot d_{prop}$.

Problem 1.2 — Protocol Layers

- List the five layers of the Internet protocol stack, from top to bottom.
- A web browser sends an HTTP GET request. At which layer does each of the following operations occur: (i) breaking the request into segments, (ii) adding a source/destination IP address, (iii) determining the next-hop MAC address?
- Explain the concept of *encapsulation*: what happens to data as it travels down the stack?

2 Information Theory & Entropy (Lecture 2)

Problem 2.1 — Shannon Entropy

A discrete memoryless source produces symbols from the alphabet $\{A, B, C, D\}$ with probabilities $p_A = 1/2$, $p_B = 1/4$, $p_C = 1/8$, $p_D = 1/8$.

- (a) Compute the entropy $H(X)$.
- (b) What would the entropy be if all four symbols were equally likely?
- (c) Explain in one sentence what entropy measures in the context of data compression.

Hint

$H(X) = -\sum p_i \log_2 p_i$. For equally likely symbols: $H = \log_2 M$.

Problem 2.2 — Multimedia Data Rates

- (a) Compute the *uncompressed* data rate for stereo CD-quality audio: 44.1 kHz sample rate, 16 bits per sample, 2 channels. Express in Mb/s.
- (b) A streaming service sends 1080p video at 8 Mb/s. The uncompressed rate for 1080p at 30 fps is approximately 1.5 Gb/s. What is the compression ratio?
- (c) Why is it impractical to stream uncompressed video over a typical home internet connection?

Problem 2.3 — Units and Prefixes

- (a) A hard drive is advertised as “2 TB” using SI (decimal) prefixes. How many bytes is this?
- (b) An operating system reports the same drive as 1.82 TiB (tebibytes). Are these consistent? Explain.
- (c) A network link operates at 1 Gb/s. How long does it take to transmit 1 GiB of data? (Hint: be careful with decimal vs. binary.)

3 Physical Layer: Guided Media (Lecture 3)

Problem 3.1 — Attenuation and dB

A signal of power $P_{in} = 10$ mW is applied to a 5 km fiber-optic cable. The cable has an attenuation of 0.3 dB/km.

- (a) What is the total attenuation in dB over 5 km?
- (b) What is the output power P_{out} in mW?
- (c) Express the input power in dBm.

Hint

Loss [dB] = $\alpha \cdot d$. $P[\text{dBm}] = 10 \log_{10}(P[\text{mW}]/1 \text{ mW})$. To recover P_{out} : $P_{out}[\text{dBm}] = P_{in}[\text{dBm}] - \text{Loss}[\text{dB}]$, then convert back.

Problem 3.2 — Shannon-Hartley Theorem

A digital subscriber line (DSL) channel has a bandwidth of 1 MHz and an SNR of 40 dB.

- (a) Convert the SNR from dB to a linear ratio.
- (b) Compute the Shannon channel capacity C .
- (c) A modem achieves 20 Mb/s on this channel. How does this compare to capacity, and is it physically possible?

Hint

$\text{SNR}_{\text{linear}} = 10^{\text{SNR}[\text{dB}]/10}$. $C = B \log_2(1 + \text{SNR})$.

Problem 3.3 — Digital Signaling

- (a) Describe the difference between unipolar NRZ and polar NRZ-L. Which is preferred in practice and why?
- (b) In Manchester encoding, how is a binary “1” represented? A binary “0”?
- (c) A binary signal uses polar NRZ at 100 Mb/s. What is the symbol rate (in symbols/s)? If instead 16-PAM is used at the same symbol rate, what is the bit rate?
- (d) Name one advantage and one disadvantage of 16-PAM compared to binary NRZ.

4 Wireless Physical Layer (Lecture 4)

Problem 4.1 — Free-Space Path Loss and Link Budget

A WiFi access point transmits at $P_t = 20$ dBm with antenna gain $G_t = 2$ dBi. A laptop at distance $d = 50$ m has antenna gain $G_r = 0$ dBi. The carrier frequency is $f = 2.4$ GHz.

- Compute the free-space path loss L_{FSPL} in dB.
- Compute the received power P_r in dBm.
- The minimum sensitivity of the laptop's receiver is -80 dBm. What is the link margin?
- If the AP is moved to $d = 200$ m (same frequency), by how many dB does L_{FSPL} increase?

Hint

$L_{FSPL}[\text{dB}] = 32.45 + 20 \log_{10}(f_{\text{MHz}}) + 20 \log_{10}(d_{\text{km}})$. Link budget: $P_r = P_t + G_t + G_r - L_{FSPL}$ (all in dB/dBm/dBi). Path loss increases by $20 \log_{10}(d_2/d_1)$ dB when distance changes.

Problem 4.2 — Modulation and QAM

- How many bits per symbol does QPSK carry? 16-QAM? 64-QAM?
- A 64-QAM symbol is transmitted at 10 Msymbols/s. What is the bit rate?
- WiFi uses *adaptive* modulation. When a user moves far from an access point, the system switches from 256-QAM to BPSK. Why?
- Sketch the 4-symbol constellation for QPSK (label the I and Q axes, and show the four symbol points).

Problem 4.3 — Cellular Frequency Reuse

A cellular network uses a reuse factor of $N = 7$ and a cell radius $R = 500$ m. The path-loss exponent is $n = 3.5$.

- Compute the reuse distance D .
- Estimate the carrier-to-interference ratio (C/I) using the approximation $C/I \approx (3N)^{n/2}/6$.
- If the operator switches to $N = 3$, qualitatively what happens to (i) capacity per unit area and (ii) interference?

5 Error Control Coding (Lectures 5–6)

Problem 5.1 — Hamming Distance and d_{min} Theorem

Consider the following code with four codewords of length 6:

$$\mathcal{C} = \{000000, 001110, 110100, 111010\}$$

- (a) Compute the Hamming distance between every pair of codewords.
- (b) Determine d_{min} .
- (c) What is the maximum number of errors this code can *detect only*?
- (d) What is the maximum number of errors it can *correct only*?
- (e) Can it simultaneously detect 2 errors and correct 1? Justify using the d_{min} theorem.

Hint

$d_H(x, y)$ = number of positions where x and y differ. d_{min} theorem: $e_c + e_d \leq d_{min} - 1$, $e_c \leq e_d$.

Problem 5.2 — Code Efficiency Comparison

A network link has a raw BER of $p = 10^{-5}$ (one bit error per 100,000 bits). Frames carry $k = 1000$ data bits.

- (a) What is the probability that a 1000-bit frame contains *at least one* bit error?
- (b) (Detection + ARQ) A single parity bit is added ($n = 1001$). What is the code rate? If an undetected error occurs only when there are 2 or more errors in a frame, estimate the probability of an undetected error.
- (c) (Error correction) A Hamming code uses r check bits: $n = k + r = 1000 + 10 = 1010$. What is the code rate? Compare it to the parity code.
- (d) Which scheme provides better *throughput* for this channel? Why?

Problem 5.3 — CRC by Hand

The message is $M = 1101$ and the generator polynomial is $G(x) = x^3 + x + 1$ (bit pattern: 1011).

- (a) Write out the dividend $x^3M(x)$ (append 3 zeros to M).
- (b) Perform modulo-2 long division of $x^3M(x)$ by $G(x)$ and find the remainder $R(x)$.
- (c) Write out the transmitted frame T .
- (d) Suppose the received frame is T with bit 3 (from the left, 0-indexed) flipped. Show that the remainder after dividing by $G(x)$ is non-zero.

Hint

Modulo-2 division: at each step, XOR the partial dividend with G if the leading bit is 1; otherwise XOR with 0 (bring down next bit). No borrows, no carries.

6 Multiple Access Protocols (Lecture 7)

Problem 6.1 — ALOHA Throughput

A slotted ALOHA system has $N = 10$ nodes. Each node has a backlogged packet and independently attempts to transmit in each slot with probability p .

- (a) Write an expression for the probability that exactly one node transmits in a given slot (i.e., a successful transmission).
- (b) Using the continuous approximation $S = Ge^{-G}$ where $G = Np$, find the value of p that maximises throughput for $N = 10$.
- (c) What is the maximum throughput S_{max} in slots/slot? Express as a percentage.
- (d) How does this compare to pure ALOHA maximum throughput?

Hint

For slotted ALOHA: $P(\text{success}) = Np(1 - p)^{N-1}$. In the limit: $S = Ge^{-G}$, maximised at $G = 1$, giving $S_{max} = 1/e \approx 0.368$.

Problem 6.2 — CSMA/CD Minimum Frame Size

Classic 10 Mb/s Ethernet allows a maximum segment length of 2500 m. The propagation speed on the coaxial cable is $v = 2 \times 10^8$ m/s.

- (a) What is the one-way propagation delay τ from end to end?
- (b) What is the round-trip propagation delay 2τ ?
- (c) What is the minimum frame size in *bits* required for CSMA/CD to work correctly?
- (d) Convert to bytes. Does this match the standard 64-byte Ethernet minimum?
- (e) Why does this minimum frame size not matter on modern switched Ethernet?

Problem 6.3 — Binary Exponential Backoff

Two stations A and B both attempt to transmit and collide for the first time (collision count $k = 1$).

- (a) From what range does each station independently choose a backoff value?
- (b) What is the probability they collide *again* on the next attempt?
- (c) After a *second* collision ($k = 2$), what is the backoff range?
- (d) After how many successive collisions does CSMA/CD give up? What happens then?

7 Ethernet 802.3 (Lecture 8)

Problem 7.1 — Ethernet Frame Analysis

An Ethernet frame is captured with the following field values (hex):

Dst MAC: FF:FF:FF:FF:FF:FF Src MAC: A4:C3:F0:85:AC:2D
Type/Length: 0x0806 Payload: (46 bytes)

- (a) Is this a unicast, multicast, or broadcast frame? How can you tell?
- (b) What protocol does the EtherType 0x0806 indicate?
- (c) What is the total frame size in bytes (excluding preamble and SFD)?
- (d) What is the purpose of the 7-byte preamble and 1-byte SFD?
- (e) The FCS covers which fields of the frame?

Problem 7.2 — ARP Trace

Host A (IP: 10.0.0.1, MAC: AA:AA:AA:AA:AA:AA) wants to send an IP packet to Host B (IP: 10.0.0.2, MAC: BB:BB:BB:BB:BB:BB). Both hosts are on the same subnet. Host A has an empty ARP cache.

- (a) Describe the ARP request that A sends. What is the destination MAC address of the Ethernet frame carrying this request, and why?
- (b) Host B receives the request. Describe the ARP reply it sends.
- (c) After the ARP exchange, Host A sends the IP packet. Fill in the fields of the Ethernet frame:
 - Source MAC: _____
 - Destination MAC: _____
 - EtherType: _____
- (d) Host A now wants to send a packet to Host C (IP: 192.168.5.1) on a *different* subnet. The default gateway has IP 10.0.0.254 and MAC GG:GG:GG:GG:GG:GG. What destination MAC address does A use in the Ethernet frame?

Problem 7.3 — Switch Self-Learning

Consider the network below. Switch S1 has an empty forwarding table. Three hosts are connected: Host A on port 1, Host B on port 2, Host C on port 3.

- (a) A sends a frame to B. What does S1 do? What does S1's forwarding table contain after this?
- (b) B sends a frame to A. What does S1 do? What does S1's forwarding table contain after this?
- (c) A sends a frame to C. What does S1 do?
- (d) What would happen if there were a *loop* in the switch topology (e.g., a second cable connecting the switch back to itself)? How is this problem solved in practice?

Problem 7.4 — Ethernet Generations

- (a) Match each Ethernet standard to its line code:

Standard	Line code
10BASE-T	?
100BASE-TX	?
1000BASE-T	?

Choices: Manchester encoding; 4B5B + MLT-3; 4D-PAM5

- (b) What does “BASE” signify in the naming convention?
- (c) Why does 1000BASE-T require *four* wire pairs, while 100BASE-TX uses only two?